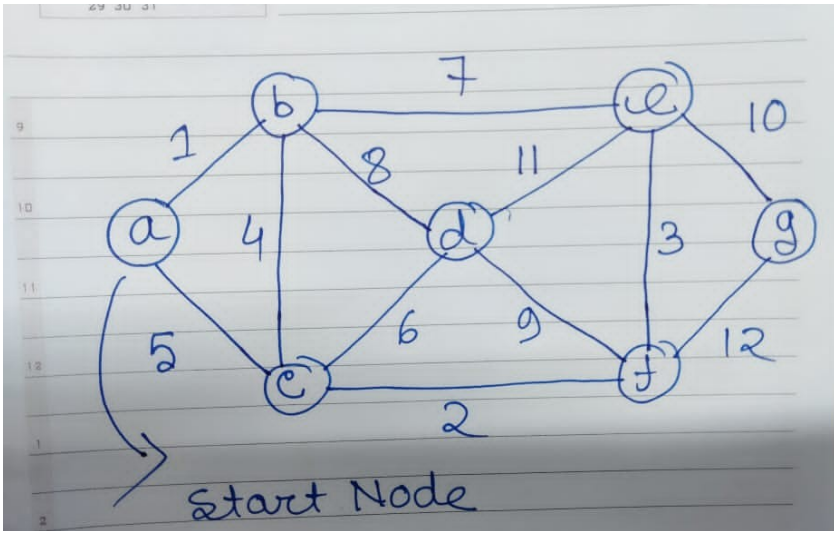
Q13.Find the cost of MST using given Graph



Ans:-

<u>Code:-</u>

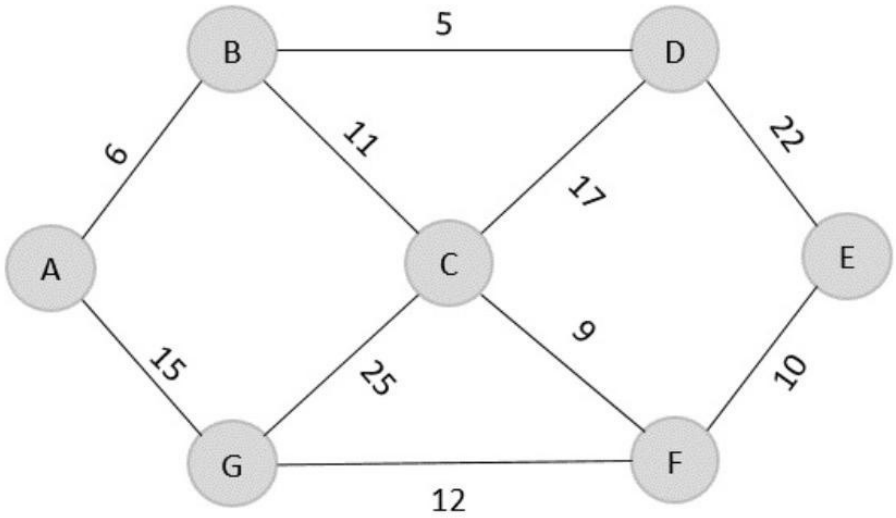
```
#include <stdio.h>

#define u 999

int main() {
    int n = 7;
    int cost[7][7] = {
        {0,1,5,u,u,u,u},
        {1,0,4,8,7,u,u},
        {5,4,0,6,u,2,u},
        {u,8,6,0,11,9,u},
        {u,7,u,11,0,3,10},
        {u,u,2,9,3,0,12},
        {u,u,u,u,10,12,0}
    };
```

```
    int visit[7] = {0};
    int mincost = 0;
    visit[0] = 1;

    for (int e = 0; e < n - 1; e++) {
        int min = u, o = 0, v = 0;

        for (int i = 0; i < n; i++) {
            if (visit[i]) {
                for (int j = 0; j < n; j++) {
                    if (!visit[j] && cost[i][j] < min) {
                        min = cost[i][j];
                        o = i;
                        v = j;
                    }
                }
            }
        }

        visit[v] = 1;
        mincost += min;
        printf("Edge %d-%d cost = %d\n", o, v, min);
    }

    printf("Minimum cost = %d\n", mincost);
    return 0;
}
```

Output

```
devildemon@devildemon-Virtua
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ gcc prim_algo_MST.c
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ ./a.out
Edge 0-1 cost = 1
Edge 1-2 cost = 4
Edge 2-5 cost = 2
Edge 5-4 cost = 3
Edge 2-3 cost = 6
Edge 4-6 cost = 10
Minimum cost = 26
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$
```

Q14. Write a C program that uses Kruskal's algorithm to find and display the minimum spanning tree (MST) of this graph and print the total weight of the MST ?



Ans:-

<u>Code:-</u>

```
#include <stdio.h>
struct edge {
    char u, v;
    int w;
};
int parent[10];

int find(int i) {
    while (parent[i])
        i = parent[i];
    return i;
}
void uni(int i, int j) {
    parent[j] = i;
}
int main() {
    struct edge e[10] = {
```

```
        {'B','D',5}, {'A','B',6}, {'C','F',9}, {'F','E',10},
        {'B','C',11}, {'G','F',12}, {'A','G',15},
        {'C','D',17}, {'D','E',22}, {'C','G',25}
    };

    int i, count = 0, sum = 0;

    printf("Edges in the Minimum Spanning Tree:\n");

    for (i = 0; i < 10 && count < 6; i++) {
        int u = find(e[i].u - 'A');
        int v = find(e[i].v - 'A');

        if (u != v) {
            uni(u, v);
            printf("%c - %c : %d\n", e[i].u, e[i].v, e[i].w);
            sum += e[i].w;
            count++;
        }
    }

    printf("Total weight of MST = %d\n", sum);
    return 0;
}
```

Output

```
                                                    devildemon@devildemon-VirtualBox
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ gcc Kruskal_algo_MST.c
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ ./a.out
Edges in the Minimum Spanning Tree:
B - D : 5
A - B : 6
C - F : 9
F - E : 10
B - C : 11
G - F : 12
Total weight of MST = 53
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$
```

Q15. A set of jobs is given where each job takes one unit of time to complete. Each job has a deadline and an associated profit. The objective is to schedule the jobs in such a way that the total profit is maximized, subject to the constraint that a job must be completed on or before its deadline.

The details of the jobs are as follows:

Jobs	A_1	A_2	A_3	A_4	A_5
Deadlines	2	1	3	2	1
Profit	60	100	20	40	20

Write a C program and display the selected job sequence to with deadlines algorithm using the Greedy approach.Also calculate the maximum profit obtained?

Ans:-

<u>Code:-</u>

```
#include <stdio.h>
struct job {
    char id;
    int deadline;
    int profit;
};
int main() {
```

```
struct job j[5] = {
    {'A',2,60},
    {'B',1,100},
    {'C',3,20},
    {'D',2,40},
    {'E',1,20}
};

int n = 5, max = 0;
for (int i = 0; i < n - 1; i++) {
    for (int k = i + 1; k < n; k++) {
        if (j[i].profit < j[k].profit) {
            struct job temp = j[i];
            j[i] = j[k];
            j[k] = temp;
        }
    }
}
for (int i = 0; i < n; i++)
    if (j[i].deadline > max)
        max = j[i].deadline;

char slot[max];
for (int i = 0; i < max; i++)
    slot[i] = '-';
```

```
    int totalProfit = 0;
    for (int i = 0; i < n; i++) {
        for (int k = j[i].deadline - 1; k >= 0; k--) {
            if (slot[k] == '-') {
                slot[k] = j[i].id;
                totalProfit += j[i].profit;
                break;
            }
        }
    }

    printf("Job sequence: ");
    for (int i = 0; i < max; i++)
        printf("%c ", slot[i]);

    printf("\nTotal Profit = %d\n", totalProfit);

    return 0;
}
```

Output

```
devildemon@devildemon-Virt
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ gcc job_sequence.c
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ ./a.out
Job sequence: B A C
Total Profit = 180
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$
```